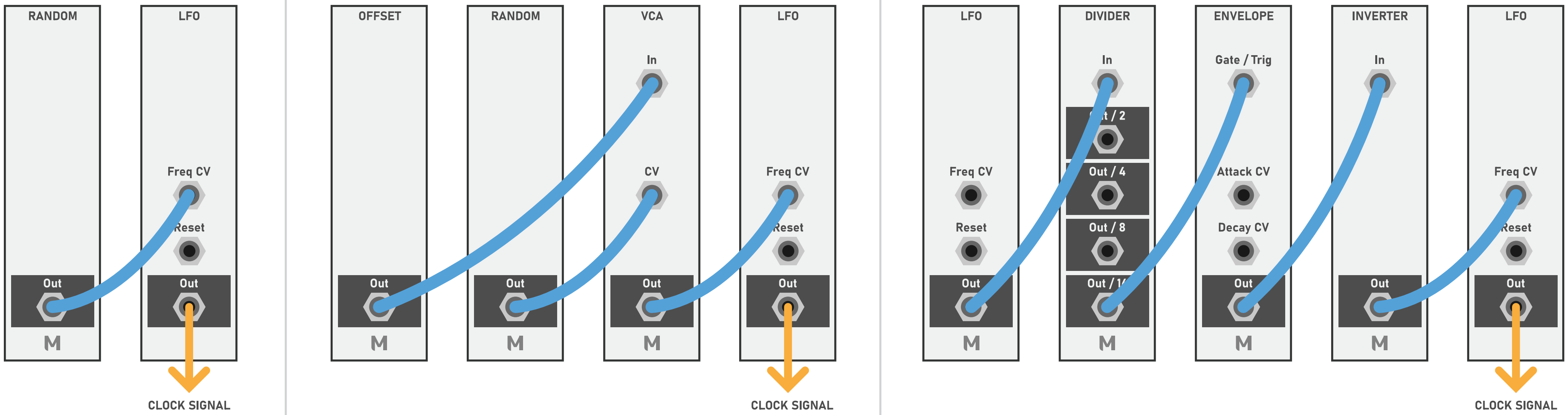
Generative
Clock

MONOTRAIL
TECH TALK

Audio
Control Voltage
Trigger / Gate



(a) (b) (b)

a - In the modular world we can use pretty much anything for a clock. A voltage controlled LFO is my favorite. You can have it run steady if you like, but also influence the clock speed with other sources. For example, have it slightly drift with a subtle random voltage.

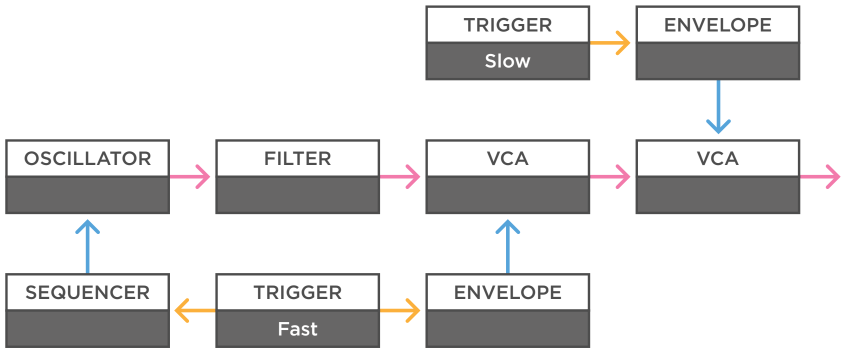
b - You can also create a steady clock that only sporadically speeds up or slows down. One way to do so is to send an offset voltage to a VCA and have that opened with a random voltage. In this case the VCA will only open, and thus allow the offset to go on and speed up the clock, when the random voltage is positive.

c - You can also use a simple attack decay envelope, triggered by a clock division or random trigger to speed up the clock every now and then. If you want to slow down the clock, just invert the signal before going on to the LFO.

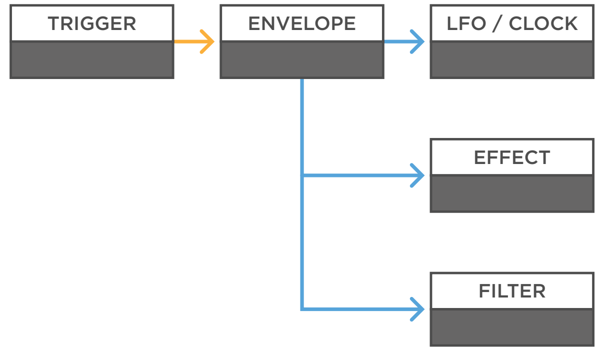
Generative
Triggers

MONOTRAIL
TECH TALK

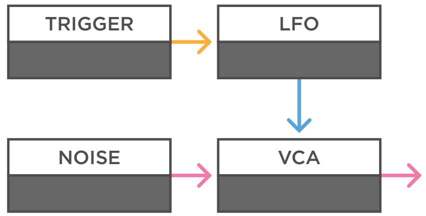
Audio
Control Voltage
Trigger / Gate



(a)



(b)



(c)

a - You can use a trigger to fire off an attack decay envelope, and send it to open a VCA, allowing a certain sound to come through. For example, you can use a slow clock division and an envelope with a long attack and decay time to allow a fast melodic voice to fade in and out every now and then.

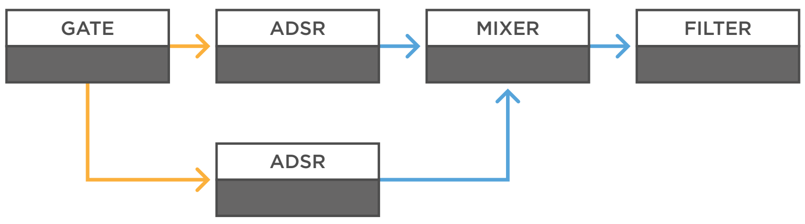
b - Other ideas are using a triggered envelope to speed up or slow down the clock speed. Send it to effects modules to have a delay or reverb swell in and out. You can also change the waveshape or open up a filter in a patch, pretty much anything works.

c - Using triggers to reset LFOs can create more interesting LFO shapes. You can use these to open a VCA allowing a more percussive noise sound to come through. Or to create interesting rhythmic modulation in a synth voice.

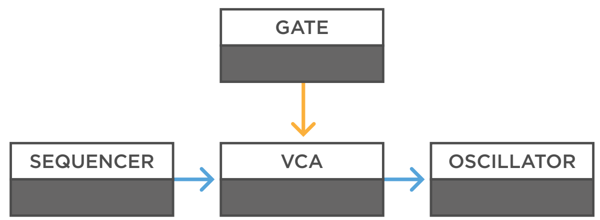
Generative

Gates

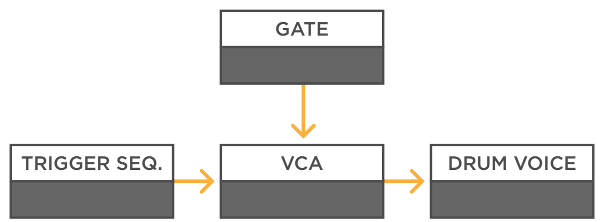
Audio
Control Voltage
Trigger / Gate



(a)



(b)



(c)

a - Gates can be used for longer effects. Send a gate to an ADSR, or even two ADSRs mixed together, if you like more creative shapes. And send that mixture to a parameter in a synth voice, like a filter or wave-folder to change the character of a sound for a sustained period.

b - You can use a gate itself to open a VCA and let something through. That can be another sound again, but also just control voltages. For example, allow a sequencer to pass on to an oscillator, turning a steady drone into a melodic part.

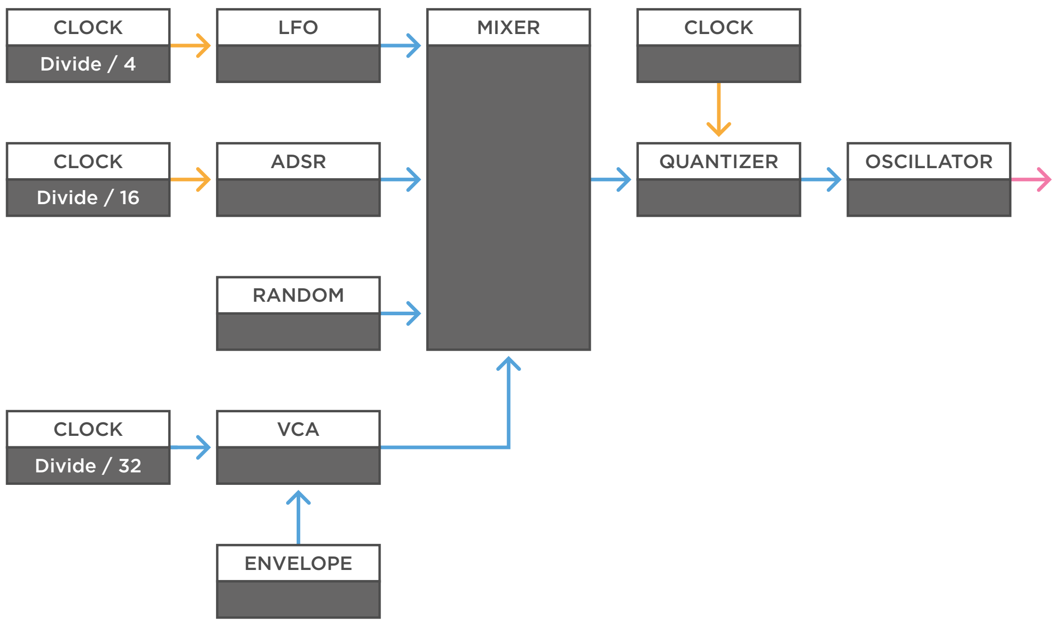
c - Or open a VCA to allow a trigger pattern to go through to a percussive module, effectively starting a drum sequence.

Generative

Generating Melodies

Audio
Control Voltage
Trigger / Gate

MONOTRAIL
TECH TALK



(a)

a - In this example, we feed a synced LFO, ADSR, and random voltage to a mixer, and then a quantizer to create interesting melodies. With some manual tweaking of the mixer, you can morph between different repeating melodies, or add some randomness to it. If you use different clock divisions to trigger or gate the signals you use, you create loops with different lengths, which can be fun.

You can add a gate from a slow clock division to the mix before the quantizer, to subtly bring the whole melody up or down for a few bars. You can for example create movement by running the gate through its own VCA and modulate it with a very slow rising envelope, to have the melody rise over time.

Generative

Generating Melodies

Audio
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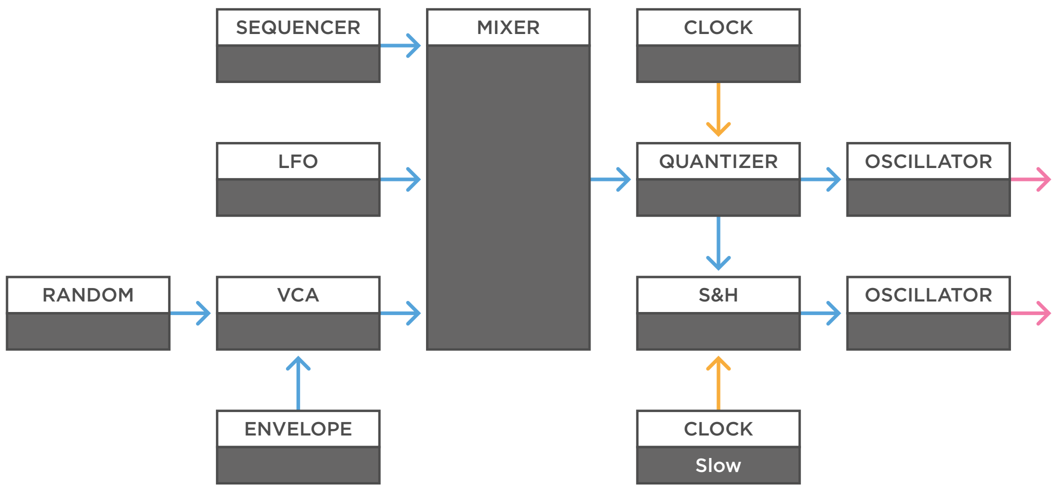

(a)

a - To create variations, you can also use a sequencer as your main source while mixing the output with other voltages. Mix them with a subtle LFO, or a smooth random voltage for example. By controlling the amount of additional signals, you can go from maintaining your original sequence with just a single note a bit higher or lower every now and then, to complete randomness.

You can have any modulation happen every now and then, by sending it through a VCA first, and have that opened with a triggered attack decay envelope or gate, like we did with the random voltage in this example.

In any case, if you modulate the output of a quantized sequencer in any way, you will have to run it through a quantizer again after modulation. But once you have a 1v/oct signal going on, you can create variations of that signal by running it through a sample & hold and triggering it with a slower speed.